

Oct 20, 2024 (Due: 08:00 Oct 27, 2025)

1. Suppose that a tall-skinny matrix $A \in \mathbb{R}^{m \times n}$ is upper bidiagonal (i.e., $a_{i,j} \neq 0$ only if $i - j \in \{0, -1\}$). Design an algorithm based on Givens rotations to solve the ridge regression problem

$$\min \|Ax - b\|_2^2 + \lambda \|x\|_2^2,$$

where λ is a given positive number.

2. Implement the algorithm we discussed in the lecture for solving the least squares problem with equality constraints (LSE), and test it on some randomly generated problems. Make sure your algorithm does *not* explicitly form the orthogonal/unitary factor in the elimination stage.

3. Let A be a square matrix, whose eigenvalues are $\lambda_1, \lambda_2, \dots, \lambda_n$, ordered with descending magnitudes. Suppose that $\lambda_1 = \lambda_2 = \dots = \lambda_k$ and $|\lambda_k| > |\lambda_{k+1}|$. Discuss the behavior of the power iteration applied to this matrix.

4. Discuss the behavior of the power iteration applied to matrices:

$$A = \begin{bmatrix} \lambda & 1 \\ 0 & -\lambda \end{bmatrix},$$

where $\lambda \in \mathbb{C}$ is a given nonzero constant.

5. Randomly generate a 1000×1000 matrix A with positive entries. Use the power iteration to compute $\rho(A)$. Visualize the convergence history.

6. (optional) Given $W \in \mathbb{C}^{n \times m}$, $V \in \mathbb{C}^{n \times p}$, and $d \in \mathbb{C}^n$, with $m \leq n \leq m + p$. Suppose that

$$\text{rank}(W) = m, \quad \text{rank}[W, V] = n.$$

Design an algorithm to solve the *generalized least squares (GLS) problem*

$$\min_{x \in \mathbb{C}^m, y \in \mathbb{C}^p} \|y\|_2 \quad \text{s.t.} \quad Wx + Vy = d.$$

Derive augmented systems of this problem.

7. (optional) Implement the PageRank algorithm discussed in the lecture and test it on some small-scale datasets.

Remark: There are several reasonably small datasets available online.